idea is that when tiny quantum jitters are smeared across space, they make it slightly hotter in one region and slightly cooler in another (photons received from a slightly denser region expend more energy overcoming the slightly stronger gravitational field, and hence their energy and temperature are slightly lower than those of photons received from a less dense

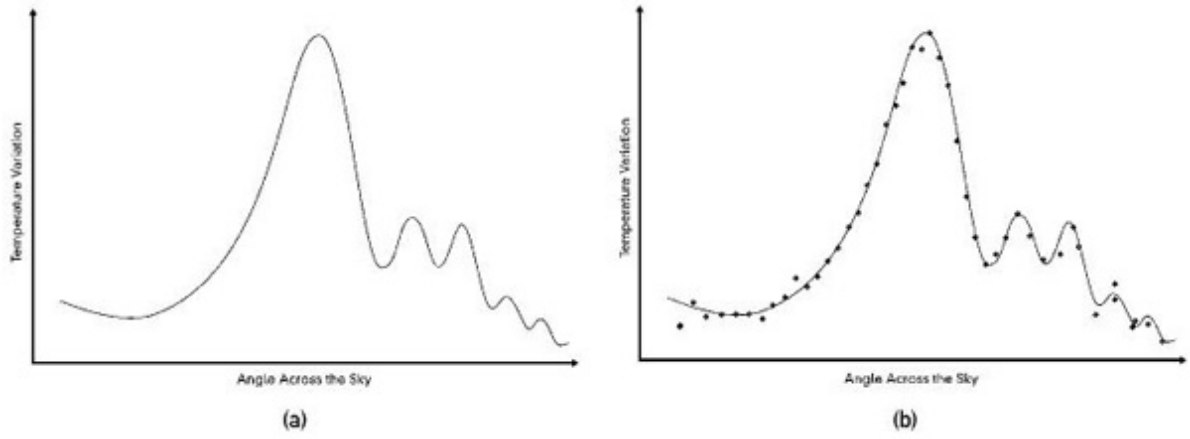



Figure 11.1 **(**a**)** Inflationary cosmology's prediction for temperature variations of the microwave background radiation from one point to another on the sky. **(**b**)** Comparison of those predictions with satellite-based observations.

region). Physicists have carried out precise calculations based on this proposal, and generated predictions for how the microwave radiation's temperature should vary from place to place across the sky, as illustrated in Figure 11.1a. (The details are not essential, but the horizontal axis is related to the angular separation of two points on the sky, and the vertical axis is related to their temperature difference.) In Figure 11.1b, these predictions are compared with satellite observations, represented by little diamonds, and as you can see there is *extraordinary agreement.*

I hope you're blown away by this concordance of theory and observation, because if not it means I've failed to convey the full wonder of the result. So, just in case, let me reemphasize what's going on here: satellite-borne telescopes have recently measured the temperature of microwave photons that have been traveling toward us, unimpeded, for nearly 14 billion years. They've found that photons arriving from different directions in space have nearly identical temperatures, differing by no more than a few ten-thousandths of a degree. Moreover, the